READ-COOP and Transkribus: cooperative approaches to sustainable and responsible digital infrastructure

Terras, Melissa
m.terras@ed.ac.uk
University of Edinburgh, United Kingdom

Anzinger, Bettina
b.anzinger@readcoop.eu
READ-COOP, Innsbruck, Austria

Muehlberger, Guenter
guenter.muehlberger@uibk.ac.at
Leopold Franzens Universität für Innsbruck, Austria

Romein, C. Annemieke
c.a.romein@utwente.nl
University of Twente, the Netherlands

Stauder, Andy
a.stauder@read.coop
READ-COOP, Innsbruck, Austria

Stauder, Florian
f.stauder@transkribus.org
READ-COOP, Innsbruck, Austria

How can we sustainably build digital scholarship infrastructures that best serve their communities, encouraging co-ownership and input into their development? Can experimenting with business models inform and propose alternative approaches to digital infrastructures which are inclusive, transparent, and responsible? This poster details the business model underpinning READ-COOP (https://readcoop.eu): an independent, jointly-owned and democratically-controlled cooperative enterprise that prioritises shared prosperity and the common needs of its members (ICA 2024). READ-COOP is a European Cooperative Society (SCE) (European Commission, n.d.) hosting Transkribus (https://transkribus.org), an Automated Text Recognition (ATR) platform that aims to unlock historical documents with AI, winning the European Union's Horizon Impact Award 2020 (EU Science & Innovation 2020). By November 2024 it had 229 members (including leading libraries, archives, and universities) from 35 countries, co-owning an infrastructure that has generated accurate transcriptions of over 100 million digital images of historical texts, with more than 300,000 registered users. The success of READ-COOP provides a blue-print for the establishment of other cooperative digital infrastructures, but also a potential alternative future towards responsible and trustworthy AI cooperatives.

READ-COOP (https://readcoop.org), and its Transkribus platform, emerged from two distinctive European Commission funded projects. TranScriptorium (2013-15, €2.4m) (European Commission 2015), led by Universitat Politècnica de València, produced the Machine Learning pipeline to generate accurate transcriptions from digital images of historical texts (Sánchez et al, 2014). The Transkribus Graphical User Interface was launched in 2015: a downloadable Java-based client programme, created by the Digitisation and Electronic Archiving group at the University of Innsbruck. The Recognition and Enrichment of Archival Documents (READ) project (2016-2019, €8.2m) (European Commission 2016), led by the University of Innsbruck, further implemented and developed the service, creating the platform as a hub for all technologies involved. By the end of the €10.6m funded period in 2019, a freely available, functioning platform for submitting and processing images was available, capable of a CER (Character Error Rate) of below 5% for handwritten text, and 1% for print material. For details of the development of the underpinning technology see Muehlberger et al (2019). To support the Transkribus infrastructure beyond this phase, it was necessary to turn it into a self-funded entity with an appropriate business model.

This poster has two aims: documenting the development of READ-COOP in its post EC funded phase (2019-2024); and to explore the nature and success of Transkribus' cooperative business model. Using Reflection In Action (Schon, 1983), to systematically capture the past 5 years history of the READ COOP we discuss the cooperative model and its suitability for AI, including the community response to being part of an AI cooperative.

Using Transkribus and READ-COOP as a case study provides an opportunity to discuss wider issues regarding the business models which underpin our digital infrastructures, including AI. Nowadays, the social, political, and economic dimensions of data are controlled by ever-fewer monopolies, intent on utilising data for capital accumulation, domination, and extraction (Hu-

berman 2022), with “our agency and autonomy becom[ing] further compromised” (Scholz 2023, 138), resulting in a “race to the bottom in terms of wages and working conditions” (Schneider 2021, 74). However, the success of the READ-COOP indicates that cooperative infrastructures can provide an alternative method to sustaining digital platforms beyond extractive “shareholder-oriented capitalism” (Cheffins 2021, 1607). We argue that cooperative digital infrastructures provide an alternative means for creating and hosting digital scholarship, digital cultural heritage, digital humanities, and AI tools, products, services, and platforms (Terras et al, forthcoming 2025). The cooperative business model supports democratic decision making while enabling revenue to improve infrastructure and services (rather than paying profits to shareholders, they are reinvested). Cooperatives, although an under-utilised legal framework and business model, hold much potential for funding specific digital infrastructures for a defined community, but also in supporting the creation of trustworthy and responsible AI (Université de Montréal 2018), in a variety of settings beyond the creative and cultural heritage sector. We argue that a cooperative approach to digital infrastructure is particularly applicable for those that have been initially developed via public funding such as research grants, provided that they have a large enough, defined, and engaged user base to support community efforts.

A change towards cooperative models for digital infrastructures would represent a significant shift in how we conceive of, implement, and sustain systems, advocating for a more democratized and equitable technology landscape. In this poster, we show that Transkribus and the READ-COOP have successfully demonstrated that potential.

This research has had research ethics approval via the READ-COOP board (October 2023) and University of Edinburgh Research Ethics processes (November 2020). The authors include the board members and executive of READ-COOP. This project has received funding from the European Union’s Horizon 2020 research and innovation programme under grant agreement No 674943 (Recognition and Enrichment of Archival Documents [READ]). This project has received funding from the European Union’s Seventh Framework Programme for research, technological development and demonstration under grant agreement No 600707 (TranScriptorium). We are thankful to the staff of READ-COOP (past and present), the members of READ-COOP, and for all interactions with READ-COOP community.

Bibliography

Cheffins, B. R. (2021). “Stop Blaming Milton Friedman!”. Washington University Law Review, Volume 98, Number 6, 2021. HeinOnline, https://heinonline.org/HOL/P?h=hein.journals/walq98&i=1629.

European Commission, n. d. “European Cooperative Society (SCE)”. https://single-market-economy.ec.europa.eu/sectors/proximity-and-social-economy/social-economy-eu/cooperatives/european-cooperative-society-sce_en [24.04.25]

European Commission, 2015. TranScriptorium, Project ID 600707, FP7-ICT-2011-9, European Commission. https://cordis.europa.eu/project/id/600707 . See https://web.archive.org/web/20121201003048/http://transcriptorium.eu/ , captured 1st December 2012.

European Commission, 2016. Recognition and Enrichment of Archival Documents (READ), Project ID 674943, H2020-EINFRA-2015-1. https://cordis.europa.eu/project/id/674943/reporting

EU Science & Innovation (2020). “TRANSKRIBUS, winner of the Horizon Impact Award 2020”. YouTube, September 23rd, 2020. https://www.youtube.com/watch?v=CL1fe3wwOaI

Huberman, J. (2022). The spirit of digital capitalism. John Wiley & Sons.

International Cooperative Alliance (2024). “Cooperative identity, values and principles” https://ica.coop/en/cooperatives/cooperative-identity.

Muehlberger, G., Seaward, L., Terras, M., Ares Oliveira, S., Bosch, V., Bryan, M., Colutto, S., Déjean, H., Diem, M., Fiel, S., Gatos, B., Greinoecker, A., Grün- ing, T., Hackl, G., Haukkovaara, V., Heyer, G., Hirvonen, L., Hodel, T., Jokinen, M., Kahle, P., Kallio, M., Kaplan, F., Kleber, F., Labahn, R., Lang, E. M., Laube, S., Leifert, G., Louloudis, G., McNicholl, R., Meunier, J. L., Michael, J., Mühlbauer, E., Philipp, N., Pratikakis, I., Puigcerver Pérez, J., Putz, H., Retsinas, G., Romero, V., Sablatnig, R., Sánchez, J. A., Schofield, P., Sfikas, G., Sieber, C., Stamatopoulos, N., Strauß, T., Terbul, T., Toselli, A. H., Ulreich, B., Villegas, M., Vidal, E., Walcher, J., Weidemann, M., Wurster, H., Zagoris, K. (2019). Transforming scholarship in the archives through handwritten text recognition: Transkribus as a case study. Journal of Documentation, 75(5):954–976, January 2019. ISSN 0022-0418. doi: 10.1108/JD-07-2018-0114. URL https://doi.org/10.1108/JD-07-2018-0114.

Sánchez, J. A., Bosch, V., Romero, V., Depuydt, K., and De Does, J. (2014). Handwritten text recognition for historical documents in the transcriptorium project. In Proceedings of the first inter-

national conference on digital access to textual cultural heritage (DATeCH '14) (pp. 111-117). https://doi.org/10.1145/2595188.2595193

Schneider, N. (2021). Enabling Community-Owned Platforms. A Proposal for a Tech New Deal. In Moore, M., and Tambini, D. (2021). Regulating Big Tech: Policy Responses to Digital Dominance. Oxford: Oxford University Press.

Scholz, R. T. (2023). Own This!: How platform cooperatives help workers build a democratic internet. London: Verso Books.

Schon, D.A. (1983) The Reflective Practitioner: How Professionals Think in Action. New York, NY: Basic Books.

Terras, M., Anzinger, B., Gooding, P., Muehlberger, G., Nockels, J., Romein, C. A., Stauder, A., Stauder, F., (Forthcoming 2025). The artificial intelligence cooperative: READ-COOP, Transkribus, and the benefits of shared community infrastructure for automated text recognition. Accepted, Open Research Europe.

Université de Montreal (2018). Montreal Declaration for a responsible development of artificial intelligence. (2017, November, 3). Announced at the conclusion of the Forum on the Socially Responsible Development of AI. https://www.montrealdeclaration-responsibleai.com/